

CS PhD Seminar Series

June 30th

| 14:30-15:30

| Room 214

Drone-Based Thermal Imaging: Principles, Applications, and Challenges

Thermal imaging has emerged as a powerful sensing technology for monitoring, inspection, and fault detection across a wide range of industrial applications. When combined with unmanned aerial vehicles (UAVs), thermal cameras enable the rapid collection of temperature data from large, inaccessible, or hazardous environments, providing significant advantages over conventional inspection methods. This seminar presents the fundamental principles of drone-based thermal imaging, including the physics of infrared radiation, thermal camera technologies, and the process of acquiring and interpreting thermal data from aerial platforms. Particular attention is given to the factors that influence the accuracy and quality of thermal measurements, such as flight altitude, viewing angle, image overlap, environmental conditions, sensor calibration, and geo-referencing. Finally, current technical challenges and future developments in drone-based thermal imaging are discussed, including multi-sensor integration, autonomous data collection, and advanced data analysis techniques.



Speaker: [Peiman Ataei](#)

Peiman Ataei is a PhD student in Computer Science at the Università degli Studi di Genova, where he obtained his MSc in Computer Science (2023). Peiman specializes in developing Android applications for DJI drone controllers, focusing on automated flight, waypoint navigation, and real-time data capture for photovoltaic inspections. His research focuses on drone-based photovoltaic inspections, integrating AI and mobile technologies to enhance renewable energy maintenance. Affiliated with JP DRONI SRL, he develops automated drone control systems under the supervision of Mr Jacopo Calla, Prof. Chessa and Prof. Delzanno.

Semi-Immersive Virtual Reality System for Motor Rehabilitation in Individuals with Rett Syndrome

Rett syndrome is a rare neurodevelopmental disorder that almost exclusively affects females, with an estimated incidence of about 1 in 10,000 girls. It is characterized by a progressive decline in motor, cognitive, and communication abilities, where eye gaze often becomes the main channel for interaction. Gross motor dysfunction in RTT significantly affects daily autonomy and quality of life. Recent studies suggest that virtual reality (VR) may enhance motor rehabilitation by increasing motivation, participation, and engagement. In this context, immersive Cave Automatic Virtual Environment (CAVE)-like systems, integrated with gaze estimation, may enable more accessible and personalized interaction through eye movements, promoting active participation in therapy.

Speaker: [Andrea De Filippis](#)

Andrea De Filippis is a second year PhD student in Computer Science at the University of Genoa-PILab (Perception&Interaction Lab). He earned his bachelor's degree in biomedical engineering and his master's degree in Bioengineering, both from the University of Genoa. His research focuses on the development and evaluation of immersive and semi-immersive virtual reality systems for motor and cognitive rehabilitation across various clinical contexts, with particular attention to Rett Syndrome and musculoskeletal disorders. He collaborates with Dialog Ausili and AIRETT (Italian Association for RETT syndrome) under the supervision of Prof. Manuela Chessa, and Prof. Danilo Pani (University of Cagliari).

